

CORPORATE NETWORKS AND PEER EFFECTS IN FIRM POLICIES

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 - ▷ Financial Investments
 - Implications for clustering of financial activity
 - ▷ Executive Compensation
 - Implications for distortions of competitive pay-scales

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 - ▷ Executive Compensation
 - Implications for distortions of competitive pay-scales
- ▶ Is there a heterogenous response to different kinds of peers?

Board interlocks are important sources of peer effects

Peer influence can be derived from:

- ▶ Network of directors: Suggestive evidence that information spreads through boardroom interactions
 - Davis & Greve 1997, Battison et.al. 2004, Khanna et. al. 2007

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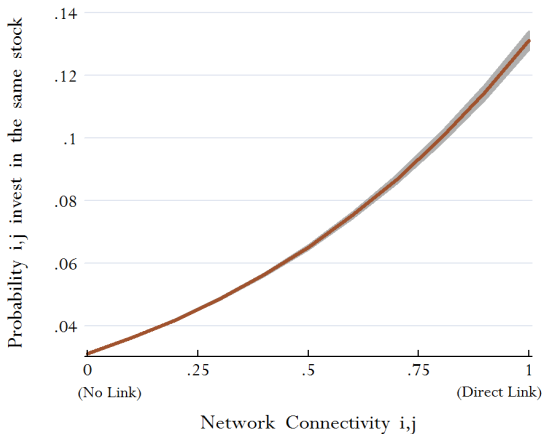
- ▶ Network of directors: Suggestive evidence that information spreads through boardroom interactions
 - Davis & Greve 1997, Battison et.al. 2004, Khanna et. al. 2007



Identification of peer *influence* is challenging

A descriptive example of network effects in investment behaviour

▷ $Pr(\text{Same Stock}_{ij}) = f(\text{Network}_{ij})$



This paper focuses on identifying the causal effect of peers

Identification summary

- ▶ Disentangle **peer effects** using network structure:
 - ▷ Endogenous peer effects: Firm i invests $\rightarrow$ Firm j invests
 - ▷ Exogenous peer effects: Firm i profits increase $\rightarrow$ Firm j invests

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- ▶ Distinguish between **peer** effects and **correlated** effects
- ▶ Non random selection into peer groups
 - ▷ Natural experiment involving death/retirement of shared directors
 - ▷ Exogenously breaks links and changes **composition** of peers

Contributions

1. Firm level peer effects arising through corporate networks

- ▶ Panel of all publicly traded firms in India from 1998-2010

Contributions

1. Firm level peer effects arising through corporate networks
 - ▶ Panel of all publicly traded firms in India from 1998-2010
2. Identification of peer effects in endogenous networks

Decisions of peers have an important influence on firm's own decisions

- ▶ Findings:
 - ▶ Strong *positive* network peer effects:
 - ▶ 1 SD $\uparrow$ in peer investment $\Rightarrow$ 0.16 SD $\uparrow$ in own firm investment
 - ▶ Social Multiplier of 1.15-1.25
 - ▶ Heterogenous Peer Effects: Non-Industry connections matter more

Introduction	Context	Identification	Data	Results	Conclusion
■					

Context

Firms regularly engage in board interlocking

- ▶ Interlocked Boards
 - ▶ **Firms are linked if they share director(s) across boards**
 - ▶ Average interlocks of firms - 9

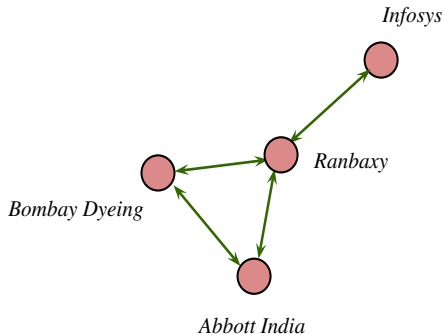
Firms regularly engage in board interlocking

- ▶ Interlocked Boards
 - ▶ **Firms are linked if they share director(s) across boards**
 - ▶ Average interlocks of firms - 9
 - ▶ Corporate governance regulations in India require:
 - ▶ Cap on max directorships (15)
 - ▶ Some restrictions on board structure: must have 50% Non-Executives

Peer groups obtained from board interlocks: Example

Company	Director	Interlocks/Peer Firms
Ranbaxy India	A Chhabra	
	B S Mehta	Sudarshan Chemical Inds. Ltd. Housing Development Finance Corpn. Ltd. Century Enka Ltd. Vinyl Chemicals India Ltd. Infosys
	B V Patel	
	Deborah Ann Henretta	
	P Agarwal	W P I L Ltd.
	R A Shah	Godfrey Phillips India Ltd. Deepak Fertilisers & Petrochemicals Corpn. Ltd. Colgate-Palmolive (India) Ltd. Bombay Dyeing & Mfg. Co. Ltd. Wockhardt Ltd. Lupin Ltd. R P G Life Sciences Ltd. Abbott India Ltd. Schrader Duncan Ltd. A C C Ltd. Century Enka Ltd. Piramal Healthcare Ltd. Clariant Chemicals (India) Ltd. Asian Paints Ltd. Pfizer Ltd. B A S F India Ltd. Atul Ltd.

Structure of interlocks gives rise to a network



Identification

Empirical Framework

- ▶ Firm i has peer group η_{it}
- ▶ Basic Estimation:

$$y_{it} = \underbrace{\beta \frac{\sum_{j \in \eta_{it}} y_{jt}}{n_{it}}}_{\text{mean outcome of peers}} + \gamma x_{it} + \underbrace{\delta \frac{\sum_{j \in \eta_{it}} x_{jt}}{n_{it}}}_{\text{mean characteristics of peers}} + u_{it}$$

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- ▶ Firm's outcome y_{it} is affected by:
 - ▶ the mean outcome of its peer group
 - ▶ its own characteristics, and
 - ▶ the mean characteristics of its peers

Network Structure

▷ Network Graph, $\mathbf{W}$

$\mathbf{W}$	1	2	3
1	0	1	0
2	1	0	1
3	0	1	0

Network Structure

- ▷ Row normalized network Graph, $\mathbf{W}$

$\mathbf{W}$	1	2	3
1	0	1	0
2	0.5	0	0.5
3	0	1	0

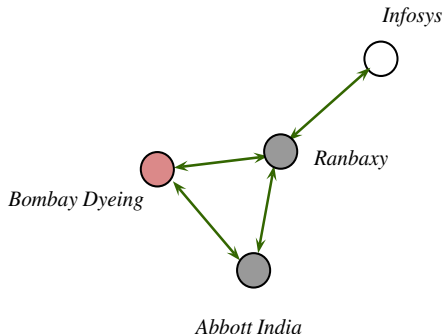
Network Structure

- ▷ Row normalized network Graph, $\mathbf{W}$

$\mathbf{W}$	1	2	3
1	0	1	0
2	0.5	0	0.5
3	0	1	0

$$y_{it} = \underbrace{\beta W_{it} y_t}_{\text{mean outcome of peers}} + \gamma x_{it} + \underbrace{\delta W_{it} x_t}_{\text{mean characteristics of peers}} + u_{it}$$

Network structure aids identification



- Intransitivity in networks:
 - ▷ Provides variations in the **magnitude** of interactions (Bramoullé, Djebbari, and Fortin, 2009)

Presence of selection & correlated effects are problematic

- ▶ OLS is biased because:

$$u_{it} = \mu_i + \nu_{it} + \varepsilon_{\eta t}$$

- ▶ μ_i Time invariant firm unobservables (selection)
- ▶ ν_{it} Time variant firm unobservables (selection)
- ▶ $\varepsilon_{\eta t}$ Correlated network unobservables

Non-Random selection can be dealt with in two ways

- Use first differences: absorbs time invariant firm unobservables

$$\Delta y_{it} = \beta \Delta W_{it} y_t + \gamma \Delta x_{it} + \delta \Delta W_{it} x_t + \Delta u_{it}$$

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$$\Delta y_{it} = \beta \Delta W_{it} y_t + \gamma \Delta x_{it} + \delta \Delta W_{it} x_t + \Delta u_{it}$$

- ▶ Local network Shocks:
 - ▷ Firm Shock: D_{it} Death/Retirement of a director in firm i

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 - ▷ Network Shock: D_{it}, D_{jt} Death/Retirement of a *shared* director b/w i and j

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$$\Delta y_{it} = \beta \Delta W_{it} y_t + \gamma \Delta x_{it} + \delta \Delta W_{it} x_t + \tau \Delta D_{it-1} + \Delta u_{it}$$

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 - ▷ Firm Shock: D_{it} Death/Retirement of a director in firm i
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- ▶ Local network Shocks:
 - ▷ Firm Shock: D_{it} Death/Retirement of a director in firm i
 - ▷ Network Shock: D_{it}, D_{jt} Death/Retirement of a *shared* director b/w i and j
- ▶ Identifying Assumptions:
 - ▷ No direct spill-over effects of firm shock on peers
 - ▷ No differential effect of death/retirement of shared directors on outcome

Death/Retirement: Firm shock and (but not necessarily) network shock

Company	Director	Interlocks/Peer Firms
Ranbaxy India	A Chhabra	
	B S Mehta	Sudarshan Chemical Inds. Ltd. Housing Development Finance Corpn. Ltd. Century Enka Ltd. Vinyl Chemicals India Ltd. Infosys
	B V Patel	
	Deborah Ann Henretta	
	P Agarwal	W P I L Ltd.
	<u>R A Shah</u>	Godfrey Phillips India Ltd. Deepak Fertilisers & Petrochemicals Corpn. Ltd. Colgate-Palmolive (India) Ltd. Bombay Dyeing & Mfg. Co. Ltd. Wockhardt Ltd. Lupin Ltd. R P G Life Sciences Ltd. Abbott India Ltd. Schrader Duncan Ltd. A C C Ltd. Century Enka Ltd. Piramal Healthcare Ltd. Clariant Chemicals (India) Ltd. Asian Paints Ltd. Pfizer Ltd. B A S F India Ltd. Atul Ltd.

Network Shocks change peer groups

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	B S Mehta	Sudarshan Chemical Inds. Ltd. Housing Development Finance Corpn. Ltd. Century Enka Ltd. Vinyl Chemicals India Ltd. Infosys
	B V Patel	
	Deborah Ann Henretta	
	P Agarwal	W P I L Ltd.
<u>Firm Shock</u> →	D_{it}	-
		-
		-
		-
		-
		-
		-
		-
		D_{it}, D_{jt}
		-
		-
		-
		-
		-
		-

Network Shock

Note: Firm and network shocks are separable

Company	Director	Interlocks/Peer Firms
Ranbaxy India	A Chhabra	
	B S Mehta	Sudarshan Chemical Inds. Ltd. Housing Development Finance Corpn. Ltd. Century Enka Ltd. Vinyl Chemicals India Ltd. Infosys
<u>Firm Shock</u> →	B V Patel	
	Deborah Ann Henretta	
	P Agarwal	W P I L Ltd.
<u>Firm Shock</u> →	D_{it}	- - - - - - - - - - - - - -
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Derive exclusion restrictions from network shock

- ▶ Objective is to generate instruments that can predict peer averages:

Networks Shock → **Loss of Peers** → **Fall in next period peer average**
proportional to:

$$W_{t-1}^D y_{t-1} \quad \text{Average outcomes of lost peers in previous time period}$$

Use outcomes and characteristics of lost peers as instruments

- ▶ Interested in Estimating:

$$\Delta y_{it} = \beta \Delta W_{it} y_t + \gamma \Delta x_{it} + \delta \Delta W_{it} x_t + \tau \Delta D_{it-1} + \Delta u_{it}$$

Use outcomes and characteristics of lost peers as instruments

- ▶ Interested in Estimating:

$$\Delta y_{it} = \beta \Delta W_{it} y_t + \gamma \Delta x_{it} + \delta \Delta W_{it} x_t + \tau \Delta D_{it-1} + \Delta u_{it}$$

- ▶ First Stage:

$$\Delta W_{it} y_t = \theta \Delta W_{it-1}^D y_{t-1} + \gamma^{fe} \Delta x_{it} + \tau^{fe} \Delta D_{it-1} + \Delta u_{it}^{fe}$$

↑ $W_{it-1}^D y_{t-1}$

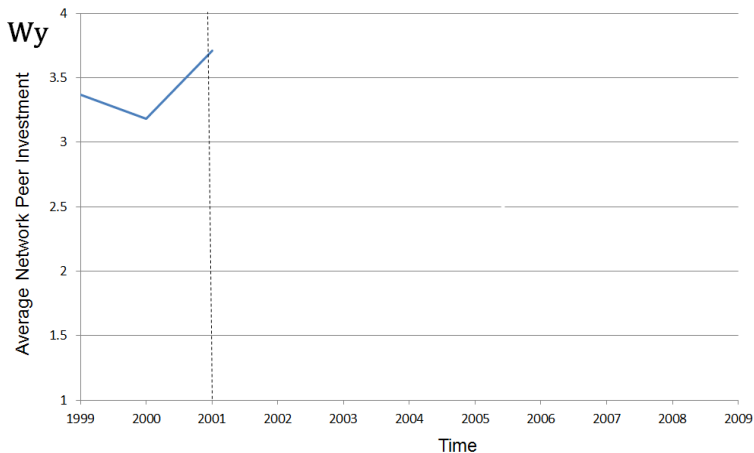
→

↓ $W_{it} y_t$

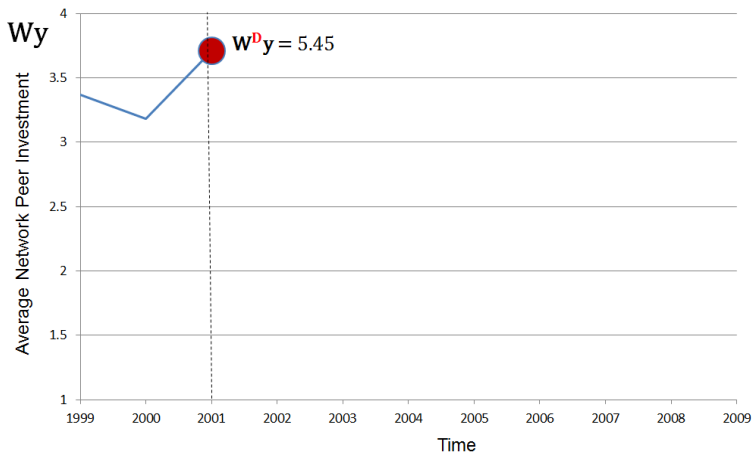
(average outcomes of lost peers)

(next period peer average)

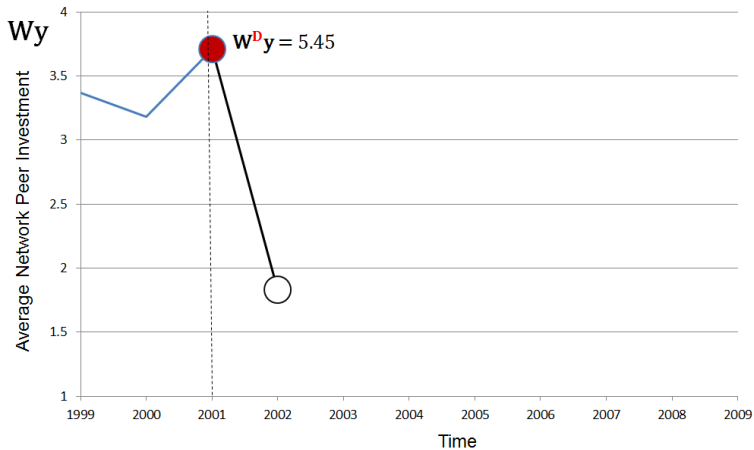
Figure plots time series of a firm's average peer investment ($W_{it}y_t$)



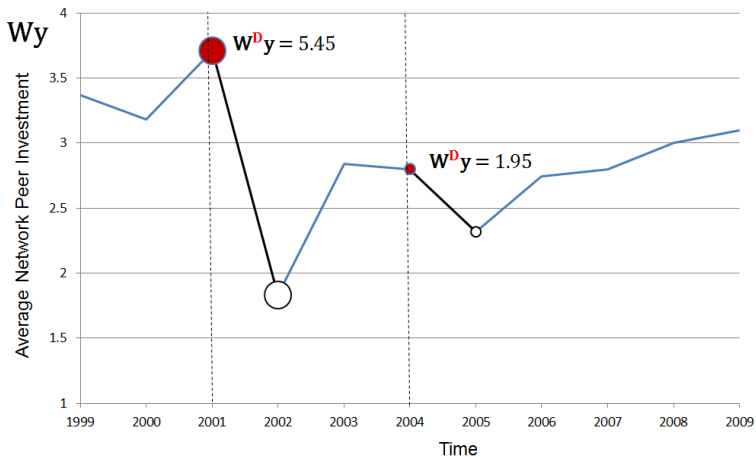
Network shock in 2001 causes peers to exit . . .



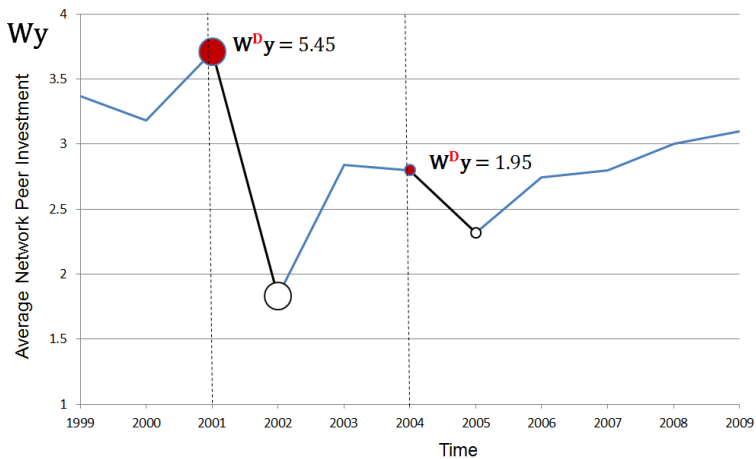
Network shock in 2001 causes peers to exit and peer average to fall in 2002



Second Network shock in 2004



$\uparrow W_{it-1}^D y_{t-1}$ (average outcomes of lost peers) $\rightarrow$ $\downarrow W_{it} y_t$ (next period peer average)



Summary

- ▶ Instrument for endogenous peer effect ($\mathbf{W}_{t\mathbf{y}_t}$) using:
 - ▶ $\mathbf{W}_{t-1}^{\mathbf{D}}\mathbf{y}_{t-1}$: Average outcomes of lost peers in previous time period
 - ▶ $\uparrow$ average outcomes of lost peers $\longrightarrow \downarrow$ next period peer average

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 - ▶ $\uparrow$ average outcomes of lost peers $\longrightarrow \downarrow$ next period peer average
- ▶ Similarly, instrument for exogenous peer effects using:
 - ▶ $\mathbf{W}_{t-1}^{\mathbf{D}}\mathbf{x}_{t-1}$: Average exogenous characteristics of lost peers

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 - ▶ $\uparrow$ average outcomes of lost peers $\longrightarrow$ $\downarrow$ next period peer average
- ▶ Similarly, instrument for exogenous peer effects using:
 - ▶ $\mathbf{W}_{t-1}^D \mathbf{x}_{t-1}$: Average exogenous characteristics of lost peers
- ▶ Use industry by time and business group by time fixed effects to capture year on common correlated shocks
- ▶ Estimate using 2SLS

Data and Results

Data Sources

- ▶ Accounting Data from PROWESS:
 - ▶ Panel of all Indian firms that are publicly listed (BSE & NSE)— from the period 1998-2010
 - ▶ Dataset classifies firms into group and non-group firms.
 - ▶ Industry Affiliation by NIC classification

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 - ▶ Dataset classifies firms into group and non-group firms.
 - ▶ Industry Affiliation by NIC classification
- ▶ Corporate Networks & Cessations Data:
 - ▶ PROWESS: Detailed information on all directors on board (matched across and within time)
 - ▶ Directors' Database: Captures reason of cessation + demographic characteristics

Results

FIRM INVESTMENT

OLS	
Network Peer Investment	0.020*** (0.005)
Lost Peer Investment (t-1)	
# Director Exits	0.003 (0.004)
Network Peer Profits	
Network Peer Assets (logs)	
Network Peer Sales (logs)	
Sales (logs)	0.040*** (0.010)
N	19644
Cragg-Donald F	

* indicates significance at 10%; ** at 5%; *** at 1%. All specifications are in first differences and include time dummies, industry by time and business group by time fixed effects. Firm level controls include assets, profits, network size and lagged investment growth.

Results

FIRM INVESTMENT

	OLS	IV: I
Network Peer Investment	0.020*** (0.005)	
Lost Peer Investment (t-1)		-0.064*** (0.006)
# Director Exits	0.003 (0.004)	-0.010+ (0.005)
Network Peer Profits		
Network Peer Assets (logs)		
Network Peer Sales (logs)		
Sales (logs)	0.040*** (0.010)	0.006 (0.015)
N	19644	19644
Cragg-Donald F		124.813

* indicates significance at 10%; ** at 5%; *** at 1%. All specifications are in first differences and include time dummies, industry by time and business group by time fixed effects. Firm level controls include assets, profits, network size and lagged investment growth.

Results

FIRM INVESTMENT

	OLS	IV: I	IV: II
Network Peer Investment	0.020*** (0.005)		0.172** (0.068)
Lost Peer Investment (t-1)		-0.064*** (0.006)	
# Director Exits	0.003 (0.004)	-0.010+ (0.005)	0.005 (0.004)
Network Peer Profits			
Network Peer Assets (logs)			
Network Peer Sales (logs)			
Sales (logs)	0.040*** (0.010)	0.006 (0.015)	0.039*** (0.011)
N	19644	19644	19644
Cragg-Donald F		124.813	

* indicates significance at 10%; ** at 5%; *** at 1%. All specifications are in first differences and include time dummies, industry by time and business group by time fixed effects. Firm level controls include assets, profits, network size and lagged investment growth.

Results

FIRM INVESTMENT

	OLS	IV: I	IV: II
Network Peer Investment	0.020*** (0.005)		0.172** (0.068)
Lost Peer Investment (t-1)		-0.064*** (0.006)	
# Director Exits	0.003 (0.004)	-0.010+ (0.005)	0.005 (0.004)
Network Peer Profits			
Network Peer Assets (logs)			
Network Peer Sales (logs)			
Sales (logs)	0.040*** (0.010)	0.006 (0.015)	0.039*** (0.011)
N	19644	19644	19644
Cragg-Donald F		124.813	

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← Multiplier = $\frac{1}{1-\beta}$ = 1.20

Results

FIRM INVESTMENT

	OLS	IV: I	IV: II	
Network Peer Investment	0.020*** (0.005)		0.172** (0.068)	← Multiplier = $\frac{1}{1-\beta} = 1.20$
Lost Peer Investment (t-1)		-0.064*** (0.006)		
# Director Exits	0.003 (0.004)	-0.010+ (0.005)	0.005 (0.004)	
Network Peer Profits				
Network Peer Assets (logs)				
Network Peer Sales (logs)				
Sales (logs)	0.040*** (0.010)	0.006 (0.015)	0.039*** (0.011)	Total Effect of Sales = 1.20×0.039
N	19644	19644	19644	
Cragg-Donald F		124.813		

* indicates significance at 10%; ** at 5%; *** at 1%. All specifications are in first differences and include time dummies, industry by time and business group by time fixed effects. Firm level controls include assets, profits, network size and lagged investment growth.

Results

FIRM INVESTMENT

	OLS	IV: I	IV: II	IV + contextual
Network Peer Investment	0.020*** (0.005)		0.172** (0.068)	0.232*** (0.088)
Lost Peer Investment (t-1)		-0.064*** (0.006)		
# Director Exits	0.003 (0.004)	-0.010 ⁺ (0.005)	0.005 (0.004)	0.006 (0.004)
Network Peer Profits				-0.000 (0.000)
Network Peer Assets (logs)				-0.054 (0.101)
Network Peer Sales (logs)				0.064 (0.108)
Sales (logs)	0.040*** (0.010)	0.006 (0.015)	0.039*** (0.011)	0.039*** (0.011)
<i>N</i>	19644	19644	19644	19644
Cragg-Donald F		124.813		

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Results

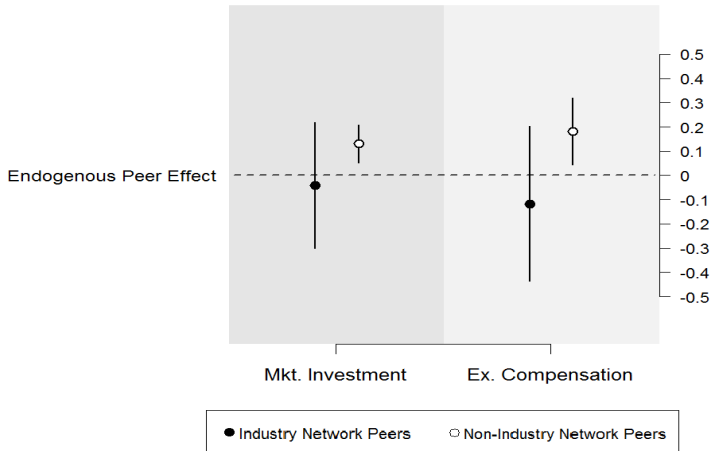
FIRM EXECUTIVE COMPENSATION

	OLS	IV: I	IV: II	IV + contextual
Network Peer Ex. Comp.	0.053*** (0.007)		0.131* (0.075)	0.167* (0.101)
Lost Peer Ex. Comp. (t-1)		-0.089*** (0.007)		
# Director Exits	-0.001 (0.001)	-0.001 (0.001)	-0.001 (0.001)	-0.001 (0.001)
Network Peer Profits				-0.01* (0.005)
Network Peer Assets (logs)				0.016 (0.026)
Network Peer Sales (logs)				0.003 (0.030)
Sales (logs)	0.005* (0.003)	-0.002 (0.003)	0.006* (0.003)	0.006* (0.003)
<i>N</i>	19644	19644	19644	19644
Cragg-Donald F		178.946		

* indicates significance at 10%; ** at 5%; *** at 1%. All specifications are in first differences and include time, industry by time and business group by time fixed effects. Firm level controls include assets, profits, network size, age and tenure of directors and lagged investment growth.

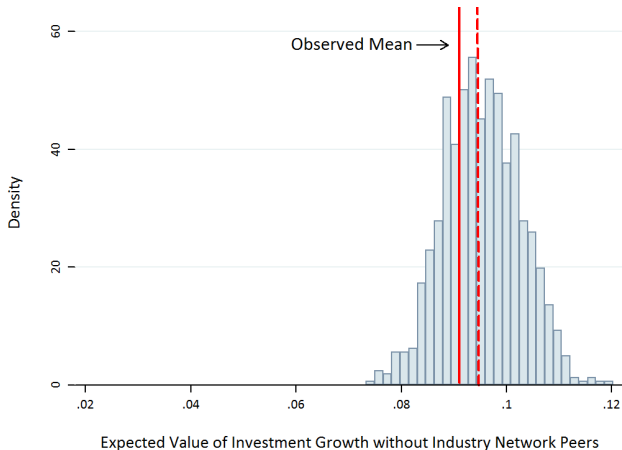
Heterogeneity in peer effects

Estimates of endogenous network peer effects - split by Industry/Non-Industry peers



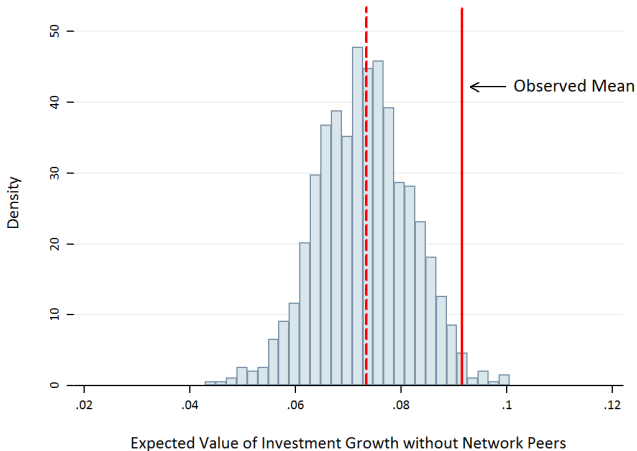
Policy: Restrict Industry Interlocks

Expected mean investment growth is not significantly different from observed mean



Policy: Restrict All Interlocks

Expected mean investment growth is (significantly) 2% below observed mean



Does the effect of firm shock differ by type of director?

Sensitivity: Direct Effect of death/retirement of *shared* director

	(1)	(2)
	Investment	Ex. Comp.
Exclusive Director Exit	0.006 (0.008)	0.0007 (0.002)
Shared Director Exit	0.010 (0.014)	-0.004 (0.004)
Peer Variables	Yes	Yes
Firm Controls	Yes	Yes
Time Fixed Effects	Yes	Yes
<i>N</i>	19644	19644

Exclusive Director Exit is a binary variable that indicates director exits exclusive of death/retirement of shared directors; *Shared Director Exit* is a binary variable that indicates director exits due to death/retirement of shared directors. Standard errors (in parentheses) are adjusted for autocorrelation and arbitrary heteroscedasticity at the network level; + indicates significance at 10%; * at 5%; ** at 1%.

Do firms strategically respond to network shocks?

Sensitivity: Probability of link replacement following a network shock

	(1)	(2)
	New Link	New Link
Link Loss (Death)	-0.002 (0.008)	-0.003 (0.009)
Firm Controls	No	Yes
Time Fixed Effects	Yes	Yes
N	19644	19644
Joint Significance of Time F.E	645.14 (0.00)	540.80 (0.00)

Conclusion

- ▶ Analyze impact of firm peer effects on corporate policies
- ▶ Exploit natural breaks in networks to identify peer effects
 - ▶ **Investment**: 1 SD $\uparrow$ in peer investment $\Rightarrow$ 0.16 SD $\uparrow$ in own firm investment
 - ▶ **Compensation**: 1 SD $\uparrow$ in peer compensation $\Rightarrow$ 0.05 SD $\uparrow$ in own firm compensation
- ▶ Effects derived mainly from non-industry network peers
- ▶ Additionally:
 - ▶ Industry peer effects: Positive Peer effects in investment and R&D
 - ▶ Magnitude of peer effects greater when considering direct + indirect links

Introduction	Context	Identification	Data	Results	Conclusion

Board structure and minutes, example

Constitution of Board & Committees

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Name	Designation	Audit Committee	Investor Grievance Committee	Compensation Committee
Mr. Kapil Kapoor	Non- Executive Chairman			
Mr. Sanjeev Bikhchandani	Founder & Executive Vice Chairman			
Mr. Hitesh Oberoi	Managing Director & Chief Executive Officer			
Mr. Ambarish Raghuvanshi	Director, Group President (Finance) & Chief Financial Officer			
Mr. Arun Duggal	Independent Director			
Mr. Ashish Gupta	Independent Director			
Mr. Saurabh Srivastava	Independent Director			
Ms. Bala Deshpande	Independent Director			
Mr. Naresh Gupta	Independent Director			

- During 2008-09, the Audit Committee undertook certain specific initiatives. This includes the following:
- Initiation of the ERP implementation project and monitoring its progress every quarter;
 - Creation of back-up certification system for CEO and CFO certificate in terms of Clause-49 of the Listing Agreement
 - Initiating and monitoring the quarterly disclosures of related party transactions.
 - Monitoring financial investments and revision of financial investment policy to suit the changing business environment
 - Made suggestions on internal verification system for effective ESOP management and administration
 - Held exclusive discussions with Internal and Statutory Auditors
 - Reviewed the Insider Trading Code and its compliance
 - Approved hedging policy for managing foreign exchange risk

B. COMPENSATION COMMITTEE As of March 31, 2009, the Compensation Committee comprises of Mr. Kapil Kapoor, Mr. Saurabh Srivastava and Ms. Bala Deshpande, Directors. The Compensation Committee held three meetings during 2008-09 on April 30, 2008, October 24, 2008, and January 22, 2009. Table 4 gives the details.

Board structure and minutes, *example*

These divisions are supported by several other businesses that focus on particular segments within the division's business domain. These include the web portal - www.naukrigulf.com that caters to the Middle-East job markets; the career counselling and guidance site - www.asknaukri.com; the professional networking site - www.brjji.com; the real estate brokerage business - www.allcheckdeals.com which is now operated through a subsidiary; and the fresher hiring site - www.firstnaukri.com.

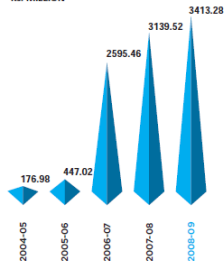
In addition to these specific business verticals, Info Edge also undertakes strategic investments in companies and start-up ventures. In 2007-08 the Company had made commitments to invest in Studyplaces Inc, USA. In 2008-09, it increased its investments by committing to take a stake in Apptect Learning Systems (Pvt.) Limited, which operates the kindergarden to class12 (K-12) www.meritnation.com, an assessment based learning portal; and Etechaces Marketing & Consulting Private Limited, which operates the insurance sales portal www.policybazaar.com.

HIGHER LIQUID ASSETS

CASH, BANK BALANCES AND LIQUID INVESTMENTS INCREASED AT A CAGR OF OVER 109.8% BETWEEN 2004-05 AND 2008-09

CASH, BANK BALANCES & INVESTMENTS

Rs. MILLION



Dealing with serial correlation

- ▶ $\mathbf{W}_{t-1}^D \mathbf{x}_{t+1}$, $\mathbf{W}_{t-1}^D \mathbf{x}_t$: Use lags of average exogenous characteristics of lost peers as instrument for endogenous peer effects
- ▶ Exogenous peer effects still identified with $\mathbf{W}_{t-1}^D \mathbf{x}_{t-1}$

Industry Peers Identification

- ▶ Construct industry peer group based on shared industry and business group firms

Industry Peers Identification

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- ▶ Use exogenous characteristics of 'peers-of-peers' to instrument for endogenous effects: (Bramoulle et. al., 2010; De Giorgi et. al., 2009)
 - ▶ Firm *i* is linked to *j* $\dashrightarrow$ same **industry**

Industry Peers Identification

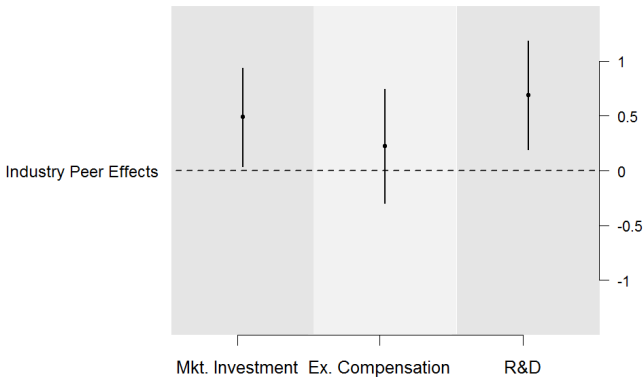
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 - ▶ Exclusion:
 - i is *not* linked to k $\dashrightarrow$ k only affects i indirectly through j

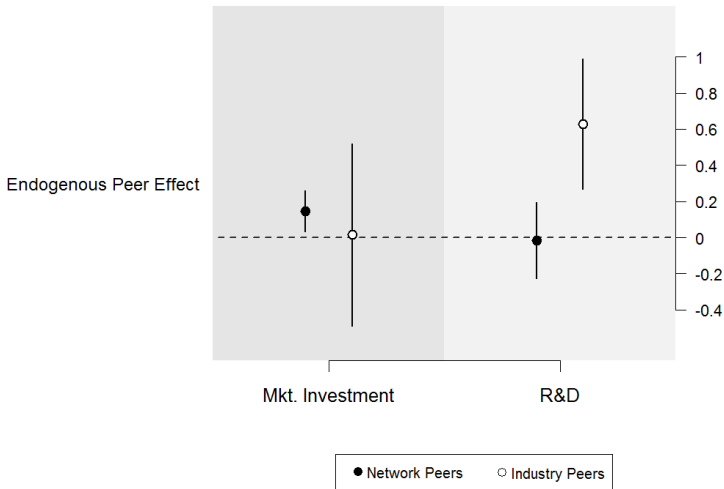
Industry peer effects

Estimates of endogenous industry peer effects



Comparing network with industry peer effects

Estimates of endogenous industry and network peer effects



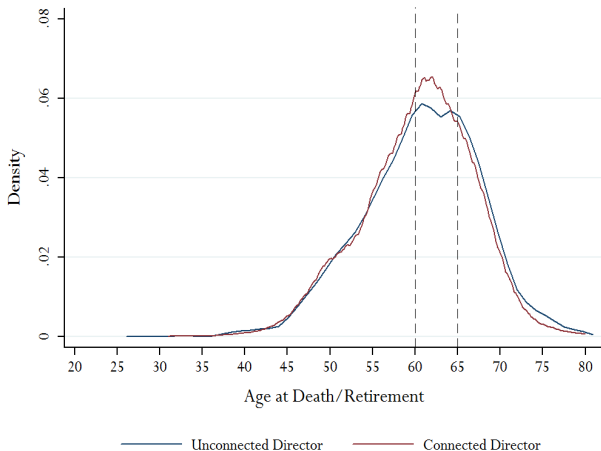
Descriptive Statistics

Variable	Mean	Std. Dev.	Median	75 th Percentile
Market Investments	1.35	1.89	0.48	2.06
Executive Compensation	0.24	0.8	0.1	0.28
Sales (log)	3.11	2.39	3.22	4.93
Profits	107.14	953.33	2.89	20.91
Assets (log)	3.97	2.03	3.76	5.26
Network Size	6.4	10.23	3	8
# Links Lost (Death)	0.29	0.95	0	0
Network Averages				
w^N Market Investments	1.5	1.5	1.23	2.44
w^N Executive Compensation	0.26	0.31	0.18	0.39
Instruments				
<i>Local Network Shocks (Reduction from death induced link loss)</i>				
w^D Market Investments	0.24	0.94	0	0
w^D Executive Compensation	0.04	0.17	0	0

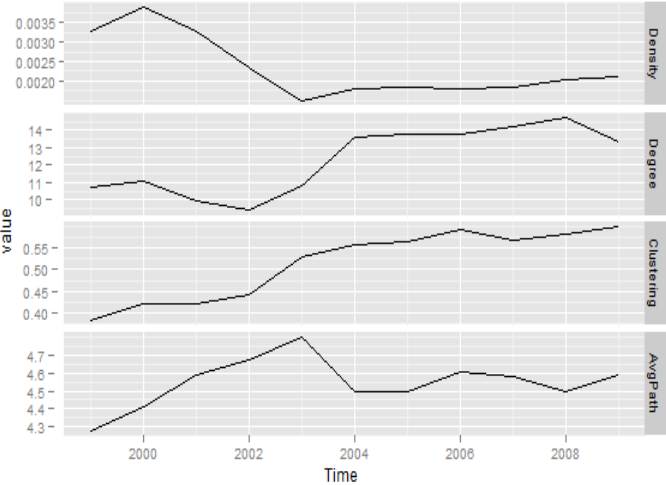


Death/retirement age similar for connected and unconnected directors

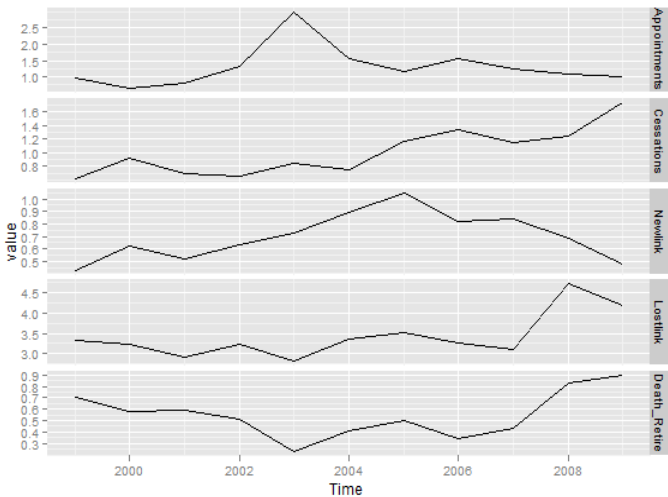
- ▷ Distribution of age at death/retirement:



Descriptive Analysis: Network Topology

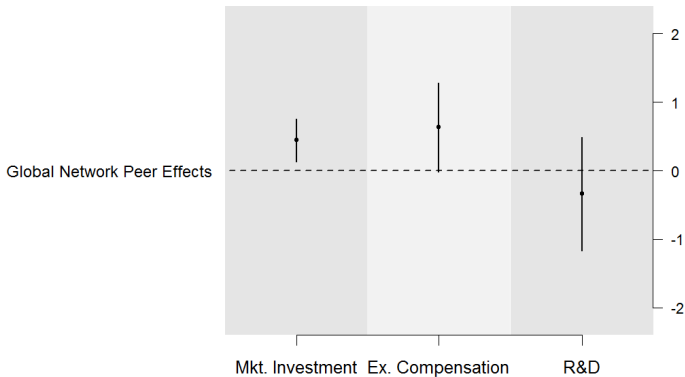


Descriptive Analysis: Network Dynamics



Global network reference group includes direct and indirect links

► Point Estimates of Global Network



Network Measurement

- ▶ Missing links of listed firms (L) to non-listed firms (NL)

